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MLA0301

Class Test-1

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Reinforcement learning

- ① Define Reinforcement learning (RL) & Explain main objective
- ② Reinforcement learning (RL) is a machine learning technique where an agent learns by interacting with an environment and receives rewards or penalties for its actions.

Main objective:
To learn the best policy that maximizes the total cumulative reward over time.

- ② what is a Markov Decision Process (MDP)? (list its components)
- ③ A Markov Decision process (MDP) is a mathematical model used to make decisions in RL.

Components:

S - States A - Actions P - Transition Probability

R - Reward function γ (gamma) - Discount factor

- ③ State the Bellman Equation for state value function

$$V(s) = \sum_a \pi(a|s) \sum_{s'} P(s'|s, a) [R + \gamma V(s')]$$

It expresses the value of state as the expected reward plus the discounted future value.

- ④ Differentiate b/w Exploration & Exploitation.

Exploration

- * Try new actions
- * Gains new knowledge
- * May give lower reward initially

Exploitation

- * Chooses the best known action
- * Maximizes immediate reward
- * Gives higher short-term reward

Q) What is multi-Armed Bandit Problem?

The multi-Armed Bandit problem is a decision making problem where an agent chooses among several actions (slot machines) to maximize total reward while balancing exploration and exploitation.

Q) What is Dynamic Programming (DP)? mention its application in RL.

Q) Dynamic Programming (DP) is a method for solving optimization problems by breaking them into smaller subproblems.

Applications in RL

- * Policy Evaluation
- * Policy Improvement
- * Value Iteration
- * Policy Iteration.

Q) What is Temporal difference (TD) learning? write the TD(0) update equation.

Temporal difference (TD) learning is an RL method that updates value estimates using the difference b/w predicted and actual rewards.

TD(0) Equation

$$V(s) = V(s) + \alpha [R + \gamma V(s') - V(s)]$$

α = Learning Rate

γ = discount factor.

Differentiate

blw

SARSA & Q-learning

SARSA

on-policy algorithm

Uses the next chosen action

Slower learning

Q-learning

off-policy algorithm

Uses the maximum Q-value

Faster learning.

Q) What is a Deep Q network (DQN)? Mention its advantages over tabular Q-learning.

A Deep Q-Network (DQN) uses a deep neural network to estimate Q-values instead of a Q-table.

Advantages:

- * Handling large state spaces
- * works with high dimensional inputs.
- * Better generalization.
- * More scalable than Q-tables.

Q) What is Double Deep Q-network (DDQN)? Why is it preferred over DQN.

Double Deep Q-network (DDQN) is an improved version of DQN reduces overestimation of Q-value by using separate networks for action selection & evaluation.

Why preferred over DQN

- * Reduces overestimation bias
- * More stable learning
- * Improves performance